28 Temperature in scanning: could thermal radiation interfere with acquisition?

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Structured light scanner Einstar Shining 3D is a low-cost scanner mainly used for videogaming production and its components count three orientated cameras with one RGB high-resolution camera for texture capturing and a projector with VCSEL technology. In previous studies, it has been already proved how the environmental light could interfere with the texture acquisition of the Einstar Shining 3D (Orsi et al., 2024); during an archaeological excavation held in Ca'Nova di Albareto, Parma, the scanner has been tested on outcropping structures, and rocks and the outputs have not been deemed as optimal due to abnormal difficulties of alignment. The misalignment problems encountered during the acquisition phase, combined with the previous experience with environmental light, brought the research team to the hypothesis of the heat waves. The reasearch team thought about how these structures were left laying under the august sun, making them reach high temperatures, faster in dissipating than the rest of ground soil. This paper investigates how thermal radiation could interfere with tools that operate using electromagnetic waves and how we have to manage the noise incrising in the acquisition due to the heat, in order to make the data usable for the FLOS pipeline of data fusion and for the drafting of the usage protocol of the scanner.